Cai Chen

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Education

Iowa State University — Ames, Iowa
Bachelor of Science in Computer Engineering — GPA: 3.2 — *Expected December 2026*

Experience

Founder and Software Engineer — February 2025 – Present
Chardle LLC — Chardle.com — *Remote*

- Built and scaled a Next.js web platform to **120,000+ monthly active users**, shipping core content, then iterating by fixing bugs and adding features such as new game modes, multiplayer, and social features.
- Cut infrastructure costs by **90%** by migrating to AWS architecture (Lambda, API Gateway, DynamoDB, S3).
- Engineered stateful WebSocket infrastructure via Cloudflare Durable Objects, achieving 1.04ms average CPU time and zero errors across **30,000+** daily real-time multiplayer requests.
- Grew monthly revenue by **220%** by negotiating and implementing an ad-tech partnership with Playwire.

Software Developer Intern — June 2026 – August 2026
W.R. Berkley — *Urbandale, Iowa*

- Shipped features to a **1.3M+ line** enterprise codebase maintained by 50+ engineers, boosting development velocity by **32%** within a dedicated 6-member agile scrum team after onboarding.
- Architected an **agentic AI workflow** with human-in-the-loop review that automated the full feature cycle (planning, implementation, testing), saving **50 hours** of development time over 2 months.
- Refactored a Node.js email microservice, consolidating duplicated logic across 10 modules and adding strict input validation and linting rules to improve longevity for a **Fortune 300 insurer**.

Associate Software Engineer (Part-Time) — August 2025 – April 2026
Todd Agriscience — *Remote*

- Architected a greenfield full-stack platform with Next.js, TypeScript, Drizzle ORM, and PostgreSQL, reducing development friction through end-to-end type safety and shared validation schemas.
- Implemented GitHub Actions CI/CD pipelines and Husky hooks, eliminating **20+ recurring build failures**.
- Co-designed a 20+ table PostgreSQL schema and full-stack data architecture integrating vector search, Sanity CMS, and S3, supporting farm, crop, livestock, and soil analysis workflows.

Software Engineering Intern — May 2025 – August 2025
Dwolla — *Des Moines, Iowa*

- Streamlined components on a fintech payment platform processing **$85B+** in annual payments, unifying 30+ fragmented UI instances, reducing legacy code redundancy and improving cross-feature consistency.
- Diagnosed and fixed a Next.js link-prefetching bug that sent excess requests to Cloudflare, preventing rate-limit IP bans for **1,200 enterprise users** from the dashboard.
- Designed and delivered **10+ features** for a payment operations dashboard using Next.js, Redux Toolkit, and TypeScript, improving fund transfer visibility and accessibility for enterprise clients.

Projects

V2.Flutr.org – Senior Design Butterfly Management App | *Drizzle ORM, PostgreSQL, Docker, DigitalOcean*

- Built a WCAG 2.1 AA-compliant multi-tenant SaaS where butterfly house staff track shipments and releases and visitors explore live species galleries, secured by 3-tier role-based access control and tenant isolation.
- Project-managed a 4-person team and co-authored a **50-page design doc** from client requirements to production, now serving **30+ staff** across **3 butterfly houses**, cutting inventory tracking time by **40%**.

Folio – HackMidwest 2026 AI Stock Research App | *Python, FastAPI, PostgreSQL, AWS Bedrock, pgvector*

- Built the data ingestion service for a stock research platform, collecting SEC Form 4 insider trades, congressional trade disclosures, and news into PostgreSQL, with Claude Haiku agent tagging and summarizing each record.

Technical Skills

Languages: TypeScript, JavaScript, Java, Python, C, Bash, SQL, HTML/CSS
Frameworks: Next.js, React.js, GraphQL, Redux Toolkit, Express.js, Node.js, FastAPI, Spring Boot, Jest, JUnit
Databases/Cloud: AWS, GCP, Docker, PostgreSQL, MongoDB, Firebase, Supabase
Tools/DevOps: Git, GitHub Actions, Jira, Linux, CI/CD, RESTful APIs